

Evaluating dynamical friction timescales in galactic cannibalism with Tree-Code simulations.

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ABSTRACT

The aim of this report is to test the validity of Chandrasekhar’s dynamical friction formula in the context of galactic cannibalism. Two Plummer distributions are initialized and evolved using a Tree-Code. The dynamical friction time is defined at the point where the angular momentum of the satellite galaxy is decreased to 10% of its initial value. The results found are mostly consistent with theory, confirming the validity of the formula.

Key words: galaxy cannibalism – dynamical friction – Tree-code

1 INTRODUCTION

The process of galactic merger is an important phenomenon, considering that the universe has formed structures through a bottom up scenario. In particular, in this report the merging time of a satellite galaxy in a more massive one is considered. This topic has been already studied with many types of codes and different galaxy initializations. In this case, for computational reasons, the number of particles used is relatively small, but still with masses resembling those of real galaxies. The analysis is checking both that the host system remains stable and that the satellite one is merging within the expected timescale due to dynamical friction.

2 THEORETICAL APPROACH

The main effects that are considered in minor mergers are dynamical friction and tidal forces acting on the satellite galaxy. In particular, regarding dynamical friction, the Chandrasekhar’s formula (1) is considered.

$$\frac{d\vec{v}_M}{dt} = -16\pi^2 G^2 M m \ln \Lambda \frac{\vec{v}_M}{v_M^3} \int_0^{v_M} dv_m v_m^2 f(v_m), \quad (1)$$

with v_M and M denoting the velocity and mass of the subject body; v_m and $f(v_m)$ representing the velocity and the distribution function of the host particles. This formula predicts the deceleration experienced by a massive object when moving in an infinite homogeneous background. Despite that, as indicated in Binney J., Galactic Dynamics (Chapter 8), it can also be applied for small bodies moving in more massive systems: so it is suitable in the case of minor merges too. As computed in Binney’s book, the theoretical friction time obtained from the integration of Equation 1 is:

$$t_{fric} = \frac{1.65}{\ln \Lambda} \frac{r_i^2 \sigma}{G M_2} = \frac{1.17}{\ln \Lambda} \frac{M_1(r_i)}{M_2} t_{cross}. \quad (2)$$

With $\ln \Lambda = \frac{r_i v_{M_2}^2}{G M_2}$, $t_{cross} = \frac{r_i}{v_{M_2}}$ and subscript 1 to indicate

quantities from the host system and 2 for the satellite ones. Equation 2 is approximately valid for any mass distribution under some conditions:

- the value of the Coulomb logarithm $\ln \Lambda$ is summed constant during the evolution since it has a slow variation with time and radius;
- the ratio μ between the perturber mass m and the interior mass of the host at the initial radius of the perturber $M(r_i)$ must satisfy $\mu \ll 1$.

So, the times obtained with Equation 2 are considered as the theoretical values to be compared with those obtained with the simulations.

Regarding tidal forces: the effect of the gravitational pull of the host on the small galaxy is treated as a tidal force. In particular, if the scale radius of the satellite galaxy a_2 is smaller than the separation R between the center of the two systems, the tidal acceleration of the satellite galaxy is given by:

$$a_t = \frac{2GM_1(r)r_s}{R^3}, \quad (3)$$

with $M(r)$ the mass of the host galaxy within the radial position of the perturber and r_s the size of the satellite.

3 NUMERICAL EXPERIMENTS

3.1 Initial conditions

3.1.1 The host galaxy

The main galaxy used is represented by the Plummer law:

$$\rho(r) = \frac{M}{\frac{4}{3}\pi a^3} \frac{1}{\left(1 + \frac{r^2}{a^2}\right)^{\frac{5}{2}}}, \quad (4)$$

a polytropic law with index $n = 5$, where M the total mass of the distribution and a the scale radius. It has an associated potential of:

$$\phi(r) = -\frac{GM}{a} \frac{1}{\sqrt{1 + \frac{r^2}{a^2}}}, \quad (5)$$

and a mass profile:

$$M(r) = M \frac{r^3}{a^3} \frac{1}{\left(1 + \frac{r^2}{a^2}\right)^{\frac{3}{2}}}. \quad (6)$$

In the simulation sets, this galaxy is represented with 5000, 10000 and 15000 particles, each of them with mass of $m = 10^6 M_\odot$. The corresponding total masses M_1 are therefore $5 \times 10^9 M_\odot$, $1 \times 10^{10} M_\odot$, $1.5 \times 10^{10} M_\odot$. The scale radius adopted is $a_1 = 10 kpc$. The mass and scale radius are chosen to resemble the real values of actual galaxies. The position of this galaxy is taken in the center of the simulation cube. The galaxy is not initialized at rest: it has a bulk velocity equal to 1% of the circular velocity of the satellite galaxy, but in the opposite direction.

3.1.2 The satellite galaxy

The satellite is modeled to mirror true dwarf galaxies. The distribution is again a Plummer profile (same as Equations 4, 5 and 6), in this case with a scale radius $a_2 = 1 kpc$ and a mass $M_2 = 0.01 M_1$, where M_1 the total mass of the host galaxy. The number of particles used are 500, 1000 and 1500; always one order of magnitude smaller than in the main galaxy. With these parameters, the central bulk of the perturber is sufficiently dense not to be immediately stripped by tidal forces. The initial position of the satellite galaxy r_i must fulfill two conditions:

- in order to use the approximated Equation 2, the mass of the host galaxy within the initial position $M_1(r_i)$ has to be way larger than M_2 (condition $\mu \ll 1$ in Section 2);
- r_i has to be sufficiently large so that the satellite galaxy is not tidally destroyed at the initial position.

Considering the first condition, it can be expressed as:

$$\mu = \frac{M_2}{M_1(r_i)} = \frac{M_2}{M_1} \frac{a^3}{r^3} \left(1 + \frac{r^2}{a^2}\right)^{\frac{3}{2}} \ll 1. \quad (7)$$

To fulfill it, r_i is chosen such that the masses within it are $0.9 M_1$ and $0.8 M_1$, obtained by inverting Equation 6. The corresponding values of μ and r for each system are reported in Table 1.

The second condition is derived from Equation 8. To avoid the disruption of the satellite system at the initial position, its self-gravity has to be larger than the tidal force from the host galaxy, i.e.:

$$a_{grav} > a_t, \quad \frac{GM_2(r)}{r^2} > \frac{2GM_1(r)r_s}{R^3}, \quad (8)$$

where $M_2(r)$ is the mass within the radius r of the considered particle of the satellite system, $M_1(r)$ the same but for the host galaxy, r_s the characteristic size of the small system on which the tidal force is acting and R the distance between the two systems. The condition (8) can be simplified considering the following assumption:

- the particle in the small galaxy is set at $r = a_2$, corresponding to $M_2(r) = 0.5 M_2$;
- r_s is assumed to be $2a_2$;
- the distance R between the two system is considered to be way larger than the size of the small galaxy, so the satellite can be treated as point-like and the $M_1(r) = M_1$.

Consequently, inequality (8) becomes:

$$\frac{0.5 M_2}{a_2^2} > \frac{4 M_1 a_2}{R^3}. \quad (9)$$

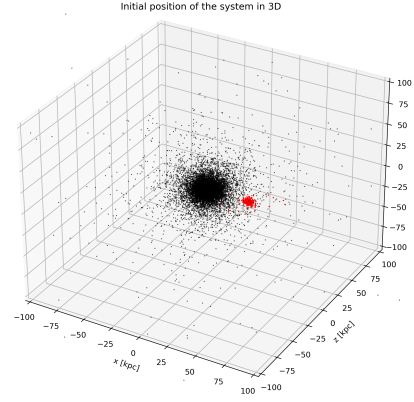


Figure 1. Example of initial conditions for a system with 10000 particles in the host (black) and 1000 in the satellite (red). All positions are in kpc .

By substituting $\frac{M_1}{M_2} = 10^2$ and $a_2 = 1 kpc$ for every system considered, it is obtained that $R \gtrsim 10 kpc$. Therefore, the initial radii r_i chosen for the satellite galaxy as $\sim 25 kpc$, for the $0.8 M_1$, and $\sim 37 kpc$, for the $0.9 M_1$, also satisfy this condition.

At the radii r , the satellite galaxy is initialized on a circular orbit with a bulk velocity equal to:

$$v_c = \sqrt{\frac{GM_1(r)}{r}}, \quad (10)$$

where $M_1(r)$ is the mass of the host galaxy enclosed within the initial position r , computed from the initialized distribution of the simulation. An example of the initial positions of the initialized systems is shown in Figure 1.

3.2 Simulation's parameters

Using a Tree-Code, the internal units are chosen so that $G = 1$. This type of code has different parameters for the integration that have to be set:

- integration time step t_{int} ;
- output time t_{out} ;
- stop time t_{stop} ;
- softening parameter ϵ ;
- accuracy angle θ .

The time between each integration step is computed from the dynamical time. In particular, the dynamical time is approximately equal to the orbital one:

$$t_{dyn} \sim t_{orb} = 2\pi \sqrt{\frac{r_i^3}{GM}}. \quad (11)$$

From this, the integration step time is obtained as $t_{int} = \frac{1}{1000} t_{dyn}$, while the output time is chosen to be $t_{out} = 100 t_{int}$. The total integration time t_{stop} for each simulation is computed from taking 1.5 times the theoretical friction time (obtained from Equation 2): $t_{stop} = 1.5 t_{fric}$. For the 0.9 simulations with 10000 and 15000 particle in the host galaxy, the stopping time is reduced to $t_{stop} = 1.3 t_{fric}$, for computational reasons. The Tree-Code also includes a softening parameter. In the definition of the force acting on a particle, instead of using only the distance between two particles, a softening term is summed in quadrature:

$$|\vec{F}_{ij}| = G \frac{m_i m_j}{|\vec{r}_i - \vec{r}_j|^2 + \epsilon^2}. \quad (12)$$

This reduce strong scatterings between two particles, effectively "softening" their interactions at very small impact parameters. The value of ϵ is computed from of the influence radius of the satellite galaxy, considered as point like. In particular:

$$R_{infl} = \frac{GM_2}{\sigma^2}, \quad (13)$$

where the velocity dispersion of the hosting Plummer is:

$$\sigma^2 = \frac{GM_1}{6a_1} \left(1 + \frac{r^2}{a^2}\right)^{-\frac{1}{2}}. \quad (14)$$

Since σ^2 is decreasing with radius, the influence radius is minimized at the center of the distribution. So, the value for ϵ is taken as a fraction of the R_{infl} at the center, therefore:

$$\epsilon = \frac{1}{50} R_{infl}(0) = \frac{M_2}{M_1} 6a_1. \quad (15)$$

The last parameter is the accuracy angle: the threshold value under which the force acting on a particle coming from a portion of the distribution can be approximate by its center of mass. In this case, $\theta = 0.1 \text{ rad}$ is adopted. All parameters are summarized in Table 1.

4 RESULTS

The simulations described above are performed with the aim of obtaining an estimate of the decay time of the satellite system.

To verify that the system behaves as expected, various quantities are analyzed. First, the stability of the host system through time is checked by performing the Lagrangian radii. In particular, since the bigger system is initialized with a bulk velocity, the trajectories of all the particles should be corrected for the motion of the center of mass. The center of mass is obtained in a recursive way: at each time step, starting from an initial aperture radius in which most of the particles are embedded (chosen as five times the position of the outermost particle at the first time step), the components x, y and z of the center of mass are calculated. From that, the aperture radius is then decreased and the new components are computed iteratively. The real value of the center of mass is taken when the variation of its norm between two subsequent values is less than 0.1%. By doing this procedure, the center of mass is less affected by unbound particles that can be scattered outside the system (it is checked that the particles lost due to larger orbits or scattering is less than 1% of the total initialized mass in all simulation). Given the evolution in time of the center of mass, the particles coordinates are corrected, i.e.: $x_i = x_i - x_{CM}$ (similarly for y and z). The radial positions are then estimated from the norm: $r_i = \sqrt{x_i^2 + y_i^2 + z_i^2}$. The Lagrangian radii are obtained from these corrected values for the particles of the host galaxy, in mass fractions between 0.1 and 0.8. As shown in Figure 5, the host system seems to be stable in each simulation.

Only the case with 5000 particle and the perturber starting at 0.9 presents an expansion in the outer region. This may be due to a stronger interaction between the satellite and the host; since the host has less particles, the interactions are fewer. So, during the merger with the other galaxy, the system reconfigures itself into a more extended structure.

Next, the behavior of the satellite galaxy is analyzed. As before, the center of mass is computed recursively from the particles positions already corrected for the motion of the bigger system. Then, the radius of the center of mass is plotted as function of time (Figure 6): the expected decay into the inner parts of the host system is present in each simulation. A note on the behavior of the satellite galaxy:

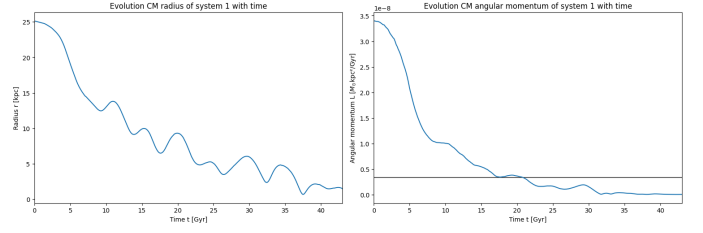


Figure 2. Evolution in time of the system with 5000 and 500 particles, with initial position at 25 kpc . Left panel: Decay of satellite's center of mass radius (in kpc) with time (in Gyr). Right panel: Decay of satellite's center of mass angular momentum (in $M_\odot \text{ kpc}^2/\text{Gyr}$) with time (in Gyr).

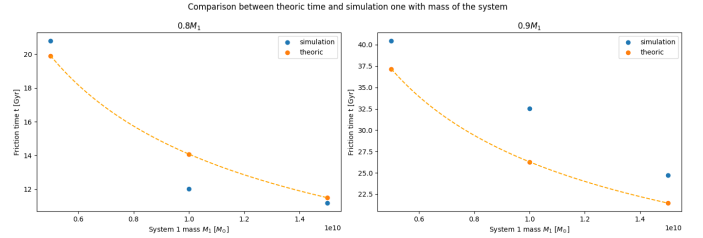


Figure 3. Comparison between the theoretical values (orange) and the simulation times (blue). The x-axis represents the mass of the host system in $M_\odot$, while the y-axis shows the time expressed in Gyr . In the left panel, the initial radius of the perturber is 25 kpc , while in the right one it is 37 kpc .

it starts as a compact system decaying in the larger one; but after a few orbits its outer region becomes tidal disrupted. The loss of some particles is not really a problem, they still decay into the bigger system and settle in a stable configuration, while the more dense core of the perturber galaxy continues to sink. Ultimately, the small Plummer system is destroyed but its particles have reached equilibrium within the bigger distribution.

To verify that also the small system remains stable over longer timescales, the first system with 500 particles is evolved till a final time of $2.5 t_{fric}$. As shown in Figure 2, the system remains in the new configuration joined with the host system.

The angular momentum of the satellite's center of mass is computed too.

To estimate the decay time, the evolution of the angular momentum is used. In particular, the friction time is defined as the first point at which the angular momentum L_{tf} becomes less than 10% of the initial value L_0 , i.e.: $L_{tf} < 0.1 L_0$ (Figure 7).

The obtained values of the time and the expected theoretical ones are in Table 2 and are compared with the theoretical ones in Figure 3.

All values found seems to be in agreement with the theoretical predictions. Only the simulations with 1000 particles in the primary system have a larger discrepancy. Specifically, the simulations times at 0.8 present a slight underestimation compared to the expected values, while those at 0.9 are overestimated. The reason behind the overestimation may be due to the lower density at bigger radii: since the friction time is proportional to the density of the distribution, smaller densities lead to longer times. Moreover, the dynamical friction force in the code is less effective in the outer regions still due to the lower number of particles. To perform a better analysis, a linear fit can be computed. Considering the quantities and approximation used, Equation 2 can be simplified to: $t_{fric} \propto \frac{1}{\sqrt{M_1(r)}}$. Thus, the

N_{p1}	N_{p2}	$r[kpc]$	$\%M_1(r)$	μ	t_{int}	t_{out}	$t_{stop}[Gyr]$	ϵ	θ
5000	500	25	0.8	0.011	1.3e-5	1.3e-3	29.9	0.01	0.1
10000	1000	25	0.8	0.011	8.8e-6	8.8e-4	21.1	0.01	0.1
15000	1500	25	0.8	0.011	7.2e-6	7.2e-4	17.2	0.01	0.1
5000	500	37	0.9	0.012	2.1e-5	2.1e-3	55.8	0.01	0.1
10000	1000	37	0.9	0.012	1.5e-5	1.5e-3	34.2	0.01	0.1
15000	1500	37	0.9	0.012	1.2e-5	1.2e-3	26.6	0.01	0.1

Table 1. Summary of the parameters used in the simulations. N_p are the number of particles in the systems. The values of the time steps are expressed in internal units.

N_{p1}	$\%M_1(r_i)$	t_t	t_s	Δt
5000	0.8	19.9	20.8	0.9
10000	0.8	14.1	12.0	2.1
15000	0.8	11.5	11.2	0.3
5000	0.9	37.1	40.5	3.3
10000	0.9	26.3	32.5	6.2
15000	0.9	21.5	24.7	3.2

Table 2. Summary of the results obtained for the friction time. N_{p1} is the number of particles in galaxy 1. t_t indicates the theoretical prediction from Chandrasekhar’s formula. t_s denotes the times obtained from the simulations. Δt is the absolute difference between the two times. All values are expressed in *Gyr*.

mass. In this report, it is also found that the effect of particles tidally destructed by the host system on the computation of the friction time is negligible. Moreover in this case, it can be performed a better analysis to determine how much mass of the satellite system can be lost before the non-validity of Chandrasekhar formula.

REFERENCES

Binney J., Galactic Dynamics

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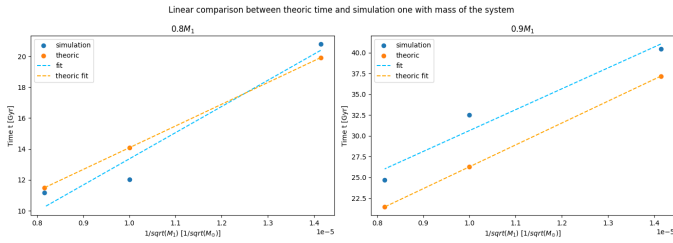


Figure 4. Comparison between the theoretical values (orange) and the simulation times (blue) after linearization. The x-axis is the inverse of the square root of the mass of the host system in $1/\sqrt{M_\odot}$, while the y-axis shows the time expressed in *Gyr*. In the left panel, the initial radius of the perturber is $25kpc$, while in the right one is $37kpc$.

substitution $x' = \frac{1}{\sqrt{x}}$ is performed (Figure 4). The obtained behavior is influenced by the central point at 1000 particles, which is underestimated in the first one and overestimated in the second; however, in general the points seems to behave as the theoretical trend. Another hypothesis on the trend of the values is considered: maybe the underestimations or overestimations may arise from differences between the real values of the circular velocity of the satellite galaxy and the one that instead computed numerically from the initialized distribution. Even if some values are slightly different than the theoretical one, a common trend is not found.

5 CONCLUSIONS

The dynamical time of a galaxy cannibalism system is computed with a Tree-Code simulation. The results seems to follow the prediction time given by Chandrasekhar’s formula. Only the case with 1000 particles in the host system is quite off. To better address the problem behind it, more simulations can be done in that range of particles and

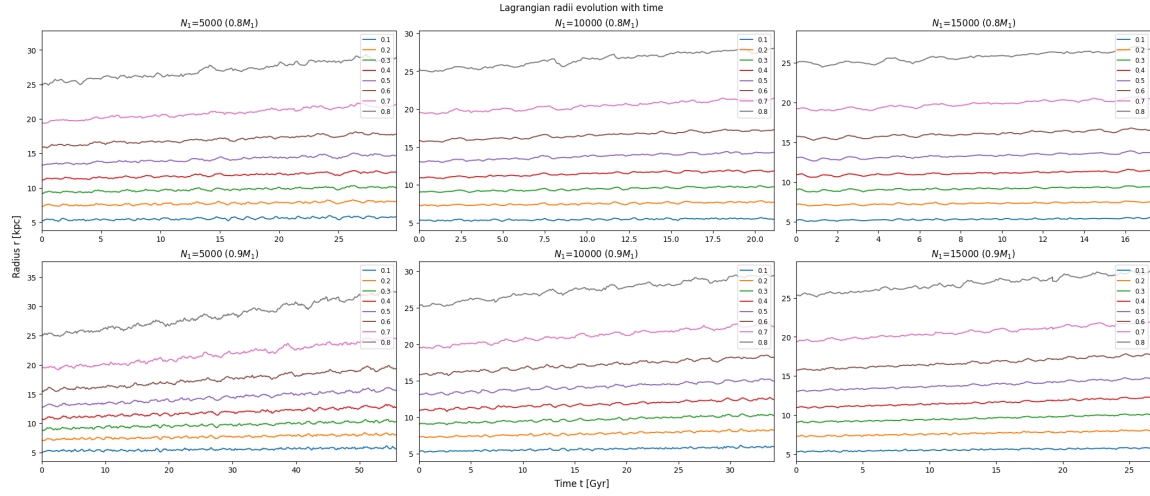


Figure 5. Evolution of Lagrangian radii with time for the different systems. Radii are in units of kpc and time in Gyr . The first corresponds to the initial radius of the satellite of $25kpc$, while the second row at $37kpc$. Each column shows the same number of particles in the host system, in order: 5000, 10000 and 15000.

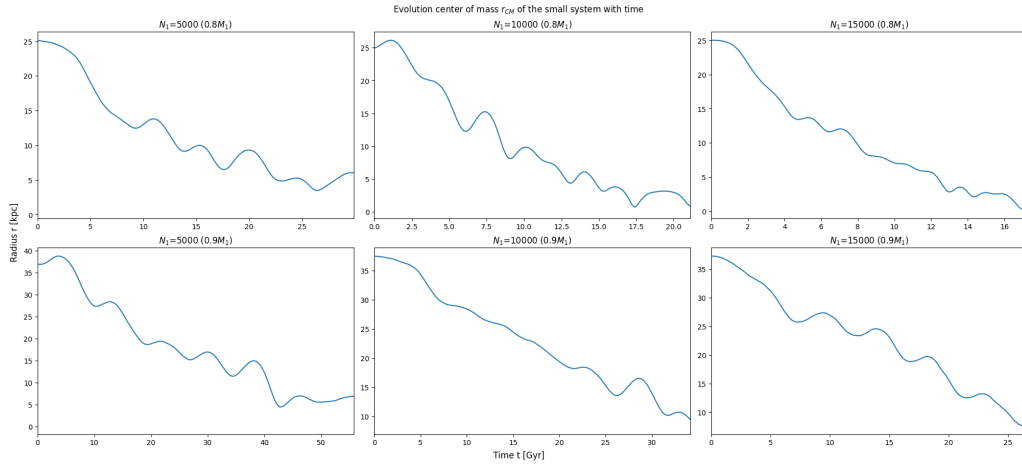


Figure 6. Evolution of the small system center of mass radius with time for the different simulations. Radii are in units of kpc and time in Gyr . The first row corresponds to the initial radius of the satellite of $25kpc$, while the second row at $37kpc$. Each column shows the same number of particles in the satellite system, in order: 500, 1000 and 1500.

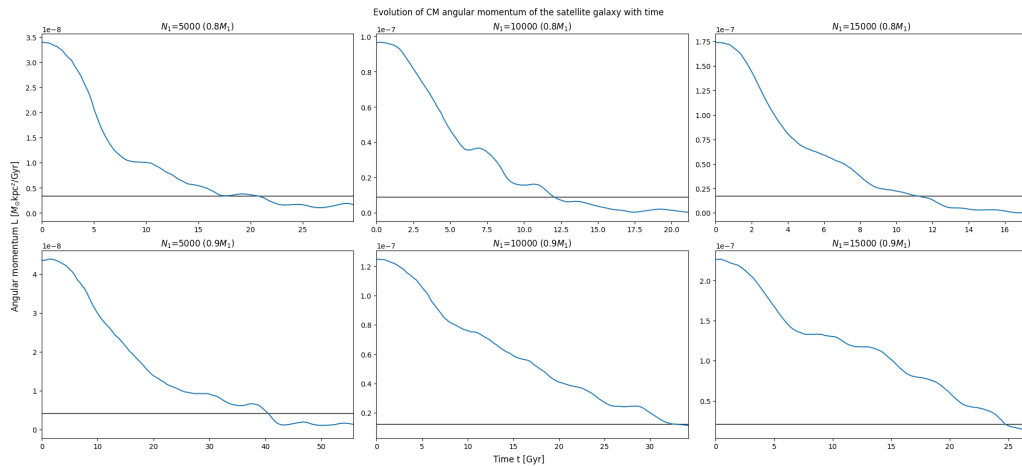


Figure 7. Evolution of the small system center of mass angular momentum with time for the different simulations. Momentum is in units of $M_{\odot}kpc^2/Gyr$ and time in Gyr . The first row corresponds to the initial radius of the satellite of $25kpc$, while the second row at $37kpc$. Each column shows the same number of particles in the satellite system, in order: 500, 1000 and 1500.